

Density + Rearranging Equations

Answer sheet · 24 marks total

Section A — Formula triangle

Top cell = m (mass) Bottom-left = ρ (density) Bottom-right = V (volume)

[1 mark each, 3 marks total]

Section B — Straight substitution

B1. $\rho = m / V = 30 / 6$ (1) $\rho = 5 \text{ kg/m}^3$ (1)

B2. $\rho = m / V = 45 / 9$ (1) $\rho = 5 \text{ g/cm}^3$ (1)

Section C — Rearrange and solve

C1. $m = \rho V$ (1) $m = 4 \times 12$ (1) $m = 48 \text{ g}$ (1)

C2. $V = m / \rho$ (1) $V = 60 / 1.5$ (1) $V = 40 \text{ cm}^3$ (1)

C3. $V = m / \rho$ (1) $V = 45 / 9$ (1) $V = 5 \text{ cm}^3$ (1)

C4. $m = \rho V$ (1) $m = 0.002 \times 2000$ (1) $m = 4 \text{ g}$ (1)

Section D — Reasoning

D1. Block B has the higher density (1), because the same mass is packed into a smaller volume — for a fixed mass, density increases as volume decreases (1).

Marking note: award method marks for a correctly rearranged equation and correct substitution even if the final answer contains an arithmetic error.